SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 3

Silver Plating + Faraday's Law

Experiment Packet

Audience: instructors and teaching assistants

Purpose: Bench instructions for instructors and TAs

Session hook: Estimate how many atoms were deposited

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 3 — Silver Plating and Faraday's Law

Hook: Calculate how many silver atoms you deposit and how thick the film is.

Learning outcomes

- Plate silver onto a metal object with controlled current and time
- Use Q = It, n = Q/F, and mass/thickness relations
- Discuss assumptions (100% efficiency, uniform coverage)
- Compare ideal Faraday prediction to visual result

Session plan (90 min)

Time	Activity	Reference
0–10 min	Hook: dull copper object — can we count the atoms we add?	lecture.md
10–25 min	Ag^{+} reduction and Faraday's law	lecture.md
25–35 min	Estimate plated surface area	experiment.md — Area
35–45 min	Set up cell with ammeter in series	experiment.md — Setup
45–65 min	Plate with timed current readings	experiment.md — Run
65–80 min	Calculate charge, moles, mass, thickness	experiment.md — Calculations
80–90 min	Discuss efficiency and side reactions	lecture.md

Key equations

```
Ag⁺(aq) + e⁻ → Ag(s)
Q = I × t
n_e = Q / F      (F = 96,485 C/mol)
n_Ag = n_e
m_Ag = n_Ag × M_Ag   (M_Ag = 107.87 g/mol)
d = m_Ag / (ρ_Ag × A)   (ρ_Ag = 10.49 g/cm³)
```

Status

- [ ] $AgNO_3$ solution prepared (0.01 M)
- [ ] Calculation worksheet ready
- [ ] Pilot plating run completed
- [ ] Waste container labeled for silver

PREPARATION

Session 3 — Preparation

Before class (instructor)

- [ ] Prepare 0.01 M $AgNO_3$ with **distilled/deionized water** (not tap); volume for all groups + 20% extra
- [ ] Pilot plate one object: record I_avg, t, appearance
- [ ] Verify 3 V packs deliver stable ~10–30 mA in this cell
- [ ] Print calculation worksheets
- [ ] Confirm silver waste disposal with school/facility protocol

$AgNO_3$ handling

- Wear gloves when preparing
- Work in ventilated area
- Label all bottles clearly
- Store away from organic materials and light if possible

Day-of setup

- [ ] Small volumes pre-measured per station (reduces spills)
- [ ] Goggles enforced at door
- [ ] Waste container visible and labeled
- [ ] Demo calculation on board with example numbers (before students start)

Timing note

Calculations take longer than expected for some groups. Have I_avg pre-computed on board as check, or provide a spreadsheet template.

Open questions / notes

- Stock $AgNO_3$ concentration on hand:
- Volume prepared (mL):
- Pilot I_avg (mA):
- Pilot plating time (s):

MATERIALS CHECKLIST

Session 3 — Materials

Per group

- [ ] 0.01 M $AgNO_3$ solution (~20–30 mL per group — prepare from stock)
- [ ] Distilled water (for rinsing)
- [ ] Object to plate: copper coin, washer, brass piece
- [ ] 3 V battery pack (preferred) or 9 V with caution
- [ ] Alligator clips
- [ ] Multimeter (current + voltage)
- [ ] Stopwatch
- [ ] Gloves, goggles, forceps
- [ ] Small beaker or jar
- [ ] Ruler (area estimation)

Optional

- [ ] Silver wire or strip (anode — replenishes Ag^+)
- [ ] Balance (for efficiency check)
- [ ] Calipers (diameter measurement)

Instructor-only

- [ ] $AgNO_3$ stock solution (instructor dilutes to 0.01 M)
- [ ] Dedicated **silver waste** container
- [ ] Spare gloves (staining risk)

Solution prep note

Target: 0.01 M $AgNO_3$ — instructor prepares before class.

Use distilled/deionized water only (tap chloride can precipitate AgCl).

Batch	Solid $AgNO_3$	Distilled water
100 mL	0.17 g	to 100 mL
250 mL	0.42 g	to 250 mL
500 mL	0.85 g	to 500 mL

Or dilute 0.1 M stock **1:10** with distilled water. Label date and concentration. ~20–30 mL per group.

Waste

- [ ] All Ag-containing liquids go to labeled waste container

EXPERIMENT PROCEDURE

Session 3 — Experiment: Silver Plating + Calculations

> **Instructors:** area pitfalls, ammeter series wiring, I_avg — instructor.md.

Silver nitrate solution — concentration and prep

Target working solution: 0.01 M $AgNO_3$ (instructor prepares before class).

Water: use distilled or deionized water only. Tap water often contains chloride, which precipitates AgCl (cloudy white) and can ruin a dilute silver bath. Rinse plated objects with distilled water too.

Batch	Solid $AgNO_3$ needed	Distilled water
100 mL of 0.01 M	**0.17 g** $AgNO_3$	Fill to 100 mL
250 mL of 0.01 M	**0.42 g** $AgNO_3$	Fill to 250 mL
500 mL of 0.01 M	**0.85 g** $AgNO_3$	Fill to 500 mL

(Molar mass $AgNO_3$ ≈ 169.9 g/mol → 0.01 mol/L × 169.9 ≈ 1.70 g/L.)

If you have a more concentrated stock

Example: dilute **0.1 M** stock **1:10** → 0.01 M
(e.g. 25 mL of 0.1 M + 225 mL **distilled** water = 250 mL of 0.01 M).

Volume per group today

20–30 mL of 0.01 M $AgNO_3$ — use the minimum that covers the object. Do not prepare large open beakers of $AgNO_3$ at student benches.

Labeling

`0.01 M AgNO₃ — Session 3 — gloves required — silver waste only`

Part A — Estimate surface area (25–35 min)

Object options

Copper coin, copper washer, cleaned brass charm, or small flat copper piece.

Methods

Coin (one face):
Measure diameter (d) in cm → (A = \pi (d/2)^2).
If both faces plate, use (A = 2\pi (d/2)^2).

Flat washer:
(A \approx \pi(R_\text{outer}^2 - R_\text{inner}^2)) (one side), or ×2 if both sides plate.

Irregular object:
Approximate immersed region as length × width (cm × cm). State the approximation in your notes.

Record

Dimension	Value (cm)	Formula used	Area A (cm^2)

Tip: A typical coin face is ~2–4 cm^2. If your A is 50 cm^2, you probably measured wrong — recheck before calculating thickness.

Part B — Setup (35–45 min)

Figure 1 — Plate the cathode object (−); measure current with the multimeter in series for Faraday calculations.

Power supply

Preferred	Acceptable
3 V battery pack	9 V with short runs and careful current checks

Target current: **~10–30 mA** (0.010–0.030 A).

Procedure

1. Put on **gloves and goggles**.

2. Pour **20–30 mL** of **0.01 M $AgNO_3$** into a small beaker or jar.
3. Clean the cathode object: sand → rinse with distilled water → dry → handle with forceps/gloves only.
4. Wire the circuit **in series**:
5. Battery (+) → anode (silver wire/strip if available; otherwise an inert conductor — note that Ag^+ will deplete)
6. Anode in solution → solution → object
7. Object (−) → multimeter **mA** input → battery (−)
   Or: battery → meter → object → solution → anode → battery.
8. Set multimeter to **DC mA** (not voltage). The meter must be **in series**, never clipped across the battery terminals alone.
9. Brief test: dip electrodes; confirm current is in the 10–30 mA range. If much higher, lift electrodes partially or use 3 V instead of 9 V.

Setup record

Item	Value
Object	
Area A (cm^2)	
$AgNO_3$ concentration	**0.01 M**
Solution volume (mL)	
Anode type	
Supply voltage	
Target plating time (s)	e.g. **300 s**
Initial current (mA)	

Part C — Timed plating run (45–65 min)

1. Start the stopwatch when a **stable** current is flowing with the object immersed.
2. Record current every **30–60 s** for the planned time (recommend **300 s = 5 min**).
3. If current drifts, keep logging — you will use the average.
4. At stop time: disconnect power first, then remove the object.
5. Rinse with distilled water; pat dry. Observe color and uniformity.

Current log

Time (s)	I (mA)	I (A) = mA ÷ 1000	Notes
0			
60			
120			

Time (s)	I (mA)	I (A) = mA ÷ 1000	Notes
180			
240			
300			

Average current:
I_avg (mA) = __ → **I_avg (A) = ____ A**
(Example: 20 mA = **0.020 A**)

Part D — Faraday calculations (65–80 min)

Use **I_avg in amperes** and total time **t in seconds**.

Constants for silver:

Quantity	Value
Faraday constant F	96,485 C/mol
Molar mass Ag	107.87 g/mol
Density ρ_Ag	10.49 g/cm^3
Electrons per Ag^+	**z = 1**

Step	Formula	Your calculation	Result
1. Charge	Q = I_avg × t		C
2. Moles e^-	n_e = Q / 96,485		mol
3. Moles Ag	n_Ag = n_e (because z = 1)		mol
4. Mass Ag	m = n_Ag × 107.87		g
5. Volume Ag	V = m / 10.49		cm^3
6. Thickness	d = V / A		cm
7. Micrometers	d_μm = d_cm × 10,000		**μm**

Worked check example (board / instructor)

I = 0.020 A, t = 300 s, A = 10 cm^2 → Q = 6.0 C → n = 6.22×10^{-5} mol → m ≈ 0.0067 g → d ≈ **0.64 μm**.

Part E — Compare to reality (80–90 min)

Discussion prompts:

1. Can you **see** a layer ~0.5–1 μm thick?
2. If current dropped, did you use **I_avg**?

3. What side reactions might steal electrons (e.g. H_2)?
4. Would doubling the time double the thickness in the ideal case?

Optional: efficiency estimate

If a balance (±0.001 g) is available:

η = (actual mass gained / calculated mass) × 100%

Safety

- **Gloves and goggles mandatory** — $AgNO_3$ stains skin and clothing permanently dark brown/black
- Small volumes only (20–30 mL per group)
- Dedicated **silver waste** container — **never** pour down the drain
- Instructor prepares/dilutes $AgNO_3$; students do not handle solid $AgNO_3$
- Wash gloves/hands protocol as directed before leaving

Student calculation sheet (duplicate for handout)

```
Given: I_avg = _____ A,  t = _____ s,  A = _____ cm²
AgNO₃ bath: 0.01 M

Q = I × t = __________ C
n_e = Q / 96485 = __________ mol
m_Ag = n_e × 107.87 = __________ g
d = m_Ag / (10.49 × A) = __________ cm = __________ µm
```

Experiment status

- [ ] 0.01 M $AgNO_3$ prepared and labeled (volume: ___ mL)
- [ ] Typical current at 3 V recorded in pilot (___ mA)
- [ ] Worksheet tested with sample numbers
- [ ] Silver waste protocol confirmed with facility

SHARED SAFETY NOTES

Syllabus Safety Notes

Apply these across all sessions. Session-specific hazards are listed in each session folder.

General rules

1. **No eating experimental materials.** Fruits, potatoes, and liquids used in experiments are no longer food.
2. **Avoid short circuits.** Batteries can heat up if directly shorted.
3. **Clean surfaces matter.** Electroplating requires clean metal surfaces.
4. **Metal salt waste should be collected.** Copper and silver solutions should not be casually poured down the drain.
5. **Water choice:** tap water is fine for most baths; use **distilled/deionized water for 0.01 M $AgNO_3$** (Session 3) so chloride does not precipitate silver.

Session-specific hazards

Session	Key hazard	Mitigation
1	Sharp nails/wires	Supervise handling; wash hands after metals
2	Copper solutions	Goggles; no ingestion; collect waste
3	Silver nitrate	Gloves + goggles; stains skin/clothing; small volumes
4	Gas evolution	No saltwater; no flames near setup; instructor-only pop test
5	Corrosion demos + electrolytic derusting	Use washing soda or baking soda only — NOT NaCl; disconnect power when not supervised

Hydrogen pop test (Session 4, instructor-only)

- Use a **tiny** volume of hydrogen only
- Perform away from the electrolysis setup
- **Never** ignite a sealed container or a mixed H_2/O_2 container